Welington Santos

Senior Software Engineer | Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **8+** years across sports commerce, e-commerce, and healthcare IT, designing and deploying production AI systems using Python and PyTorch. Experienced building ML pipelines, RAG architectures, and LLM integrations with MLOps practices, cloud deployment, and monitoring frameworks to deliver scalable, reliable AI-driven features.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Generative AI: LangChain, LlamaIndex, OpenAI API, Anthropic API, Hugging Face, LangGraph, Pinecone, ChromaDB, Weaviate, FAISS

Data & ML: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Airflow, Spark, MLflow, Weights & Biases

Backend: FastAPI, Flask, Django, Node.js, Express, gRPC, Kafka, RabbitMQ

Database: PostgreSQL, MongoDB, Redis, DynamoDB, Elasticsearch

Testing: Pytest, Jest, Playwright, Selenium

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CloudWatch, Lambda

Others: RESTful API, GraphQL, Grafana, Prometheus, Datadog, Git, GitHub, Jira, Agile

EXPERIENCE

Fanatics Oct 2023–Present

Senior Software Engineer

- Led the development of a real-time recommendation engine for the sports merchandise platform using **Python**, **PyTorch**, and **Pinecone** vector search, delivering personalized product suggestions that increased average order value by **18%**.
- Designed and deployed a **LangChain**-powered chatbot service integrated with **FastAPI** microservices to handle customer support queries, reducing average response time from 3 hours to under 5 minutes and lowering support ticket volume by **22%**.
- Spearheaded the implementation of an MLOps pipeline using **MLflow** and **GitHub Actions** to automate model training, validation, and deployment across three recommendation services, reducing time to production for new model iterations from 2 weeks to 3 days.
- Optimized model inference latency for the betting analytics service by refactoring **TensorFlow** model serving with **Docker** containerization and **Kubernetes** orchestration, cutting API response time by **35%** and supporting **15k** concurrent users.

- Collaborated with data scientists to implement **Prometheus** and **Grafana** monitoring dashboards for ML model performance, establishing alerting thresholds that caught model drift before production impact on two separate occasions.
- Integrated **OpenAI API** for generating dynamic promotional content across the platform, implementing request batching and **Redis** caching to reduce API costs by **40%** while maintaining sub-200ms response times.
- Maintained data quality across product feeds by building a **Pandas**-based validation pipeline that ran **Pytest** test suites against incoming catalog updates, preventing invalid inventory data from reaching customer-facing services.
- Refactored the feature engineering pipeline from batch processing to real-time streaming using **Kafka** and **Spark**, enabling ML models to react to customer behavior within seconds instead of daily intervals.
- Mentored two junior engineers on production ML deployment patterns, code review standards for model validation, and debugging strategies for distributed inference failures across the recommendation platform.
- Engineered a comprehensive **Datadog** observability layer across the AI services, tracking prediction latency, model accuracy drift, and API error rates, enabling the team to correlate infrastructure changes with model performance degradation.
- Implemented feature store infrastructure using **MongoDB** with **GraphQL** API to centralize feature computation and versioning, allowing data scientists to reuse validated features across 8 competing models in production.
- Coordinated a migration of legacy **Scikit-learn** models from batch scoring on **AWS Lambda** to real-time inference via **FastAPI** endpoints, supporting the expansion of personalization features to three new product categories.

Walmart Sep 2022–Oct 2023

Senior Software Engineer

- Drove the design of a product recommendation service for e-commerce using **PyTorch** embeddings and **Weaviate** vector database, delivering contextual product suggestions across Walmart.com that lifted click-through rates by **12%**.
- Built end-to-end ML pipelines for customer segmentation using **Python**, **Airflow**, and **PostgreSQL**, enabling personalized pricing strategies that increased conversion on clearance inventory by **9%** across 6 regional warehouses.
- Deployed a **LlamaIndex**-powered question-answering system over Walmart's internal documentation, integrated with **Node.js** and **FastAPI** backends, reducing time for engineers to find answers to common questions by **60%**.
- Implemented **TensorFlow** demand forecasting models optimized for supply chain operations, reducing stockouts by **7%** and excess inventory carrying costs by **11%** across grocery and apparel categories.
- Engineered model monitoring with **Weights & Biases** and **CloudWatch**, setting up automated retraining pipelines triggered by performance degradation, which prevented 3 prediction failures from reaching production.

- Scaled model serving infrastructure using **Kubernetes** and **Docker**, distributing inference across 12 nodes to handle **50k+** daily prediction requests with latency under 300ms.
- Collaborated with product and data teams to build a **Hugging Face** transformer-based NLP model for product review sentiment analysis, enabling the team to auto-flag negative customer feedback for immediate investigation.
- Optimized database queries in **PostgreSQL** for feature extraction pipelines, reducing job runtime from 4 hours to 45 minutes, which enabled faster experimentation cycles for the ML team.
- Established CI/CD practices using **GitHub Actions** for ML model validation and testing, enforcing model performance benchmarks before deployment and preventing regressions in production accuracy.

E-POT
Jun 2018–Jul 2022

Full Stack Engineer

- Built production ML inference services for healthcare diagnostics using **Python**, **TensorFlow**, and **FastAPI**, processing medical imaging data and delivering predictions within regulatory SLA requirements.
- Designed and maintained data pipelines using **Spark** and **Airflow** to aggregate patient data from multiple hospital systems, ensuring data quality through **Pandas** validation routines that ran **Pytest** integration tests.
- Implemented a RAG-based document retrieval system for medical records using **LangChain** and **ChromaDB**, allowing clinicians to query patient history in natural language with **94%** retrieval accuracy.
- Deployed ML models to **AWS** infrastructure using **Docker** containers and **Terraform** infrastructure-as-code, establishing monitoring with **Prometheus** metrics to track model performance in production.
- Developed RESTful APIs using **Flask** and **Django** to expose ML predictions to healthcare applications, handling **5k+** daily requests across 8 partner hospitals with 99.5% uptime.
- Optimized model inference performance by implementing **Redis** caching for frequently accessed predictions, reducing database load by **45%** and improving API response times by **28%**.
- Collaborated with clinical teams to gather requirements and validate ML model outputs against gold-standard diagnostics, ensuring fairness and accuracy across patient demographics.
- Mentored two junior developers on full-stack AI integration patterns, helping them ship two production features combining **React** frontends with **PyTorch** model backends.

EDUCATION

University of Tokyo
Apr 2014–Mar 2018

Bachelor of Science in Computer Science